Adaptive Boundary Consistency Learning for Open-Set Semi-Supervised Hyperspectral Image Classification

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***Abstract*—Most existing semi-supervised hyperspectral classification methods are built upon the closed-set assumption, which assumes that unlabeled samples originate exclusively from the known class space. However, this assumption is frequently invalid in real-world remote sensing scenarios. In addition, hyperspectral images (HSIs) exhibit complex spectral–spatial characteristics, facing challenges such as complex spatial boundaries, significant interclass spectral similarity, and intraclass variability. To address these issues, this article proposes an adaptive boundary consistency learning (ABCL) model for open-set semi-supervised HSI classification, which achieves low-cost and high-accuracy perception while also considering known-class classification and unknown-class rejection. First, to enhance the model's ability to represent complex land-cover boundaries and spatial structural differences, an anisotropic position transformer (APT) is designed to introduce direction-sensitive spatial priors by decoupling horizontal and vertical positional biases. Second, to address interference from unknown classes in a large number of unlabeled samples, an adaptive entropy mask (AEM) is proposed to guide consistency learning, thereby improving the discriminability of the model's features. Finally, to further optimize the feature space structure, a category recognition module (CRM) is introduced. By enhancing intraclass compactness and interclass separability, it provides a more stable representation basis for open-set discrimination. Experimental results on three public hyperspectral datasets, Indian Pines (IP), Salinas (SA), and LN02, demonstrate that with only ten labeled samples per known class, the overall accuracy (OA) of the proposed method reaches 79.56%, 88.20%, and 96.15%, respectively. Furthermore, the model exhibits stable advantages in metrics such as unknown, average accuracy (AA), and Kappa coefficient, outperforming various representative state-of-the-art methods. This study provides a low-cost, high-precision technical foundation with strong adaptability to unknown classes for target perception in HSIs.**

***Index Terms*—Consistency learning, hyperspectral image (HSI), open-set classification, semi-supervised learning.**

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I. Introduction

HYPERSPECTRAL images (HSIs) are characterized by high spectral resolution, continuous spectral bands, and rich spatial information [1], [2], [3], [4], which can provide fine-grained spectral–spatial joint representations for land-cover recognition and scene understanding. Therefore, they have important application value in fields such as environmental monitoring [5], precision agriculture [6], [7], and urban planning [8]. The goal of HSI classification is to assign each pixel to a corresponding land-cover category, such as buildings, soil, grassland, trees, rivers, and roads. As a core component of hyperspectral data processing, HSI classification plays an irreplaceable role in most hyperspectral technology applications [9], [10], [11].

The success and rapid development of deep neural networks [12] have enabled excellent performance in tasks such as image classification [13], [14], image segmentation [15], [16], and object detection [17], [18]. Inspired by this, researchers have proposed a series of deep neural network-based spectral–spatial feature extraction methods for HSIs, such as convolutional neural networks [19], [20], transformer [21], Mamba [22], and other deep models, which have significantly improved HSI classification accuracy under sufficient labeled samples. However, hyperspectral data are characterized by high spectral resolution, fine-grained categories, and complex land-cover distributions. Pixel-level annotation usually requires substantial manual effort, making it extremely difficult to obtain reliable labeled samples. Insufficiently annotated samples severely reduce the effectiveness of deep learning-based classification methods in practical applications [23].

To alleviate the performance degradation caused by the scarcity of labeled samples, semi-supervised hyperspectral classification methods attempt to jointly utilize a small number of labeled samples and a large number of unlabeled samples for training [24], and considerable progress has been

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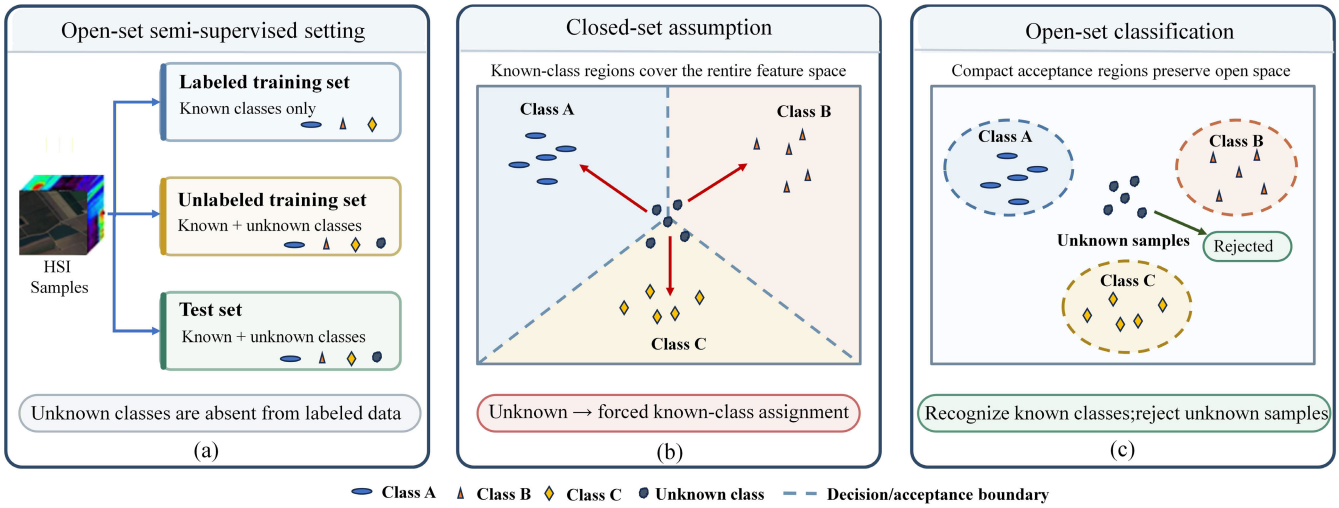


Fig. 1. Illustration of the open-set semi-supervised HSI classification setting and its difference from the closed-set assumption. (a) Labeled training set contains only known classes, whereas the unlabeled training set and test set may contain both known and unknown samples. (b) Under the closed-set assumption, the known-class decision regions cover the entire feature space. Consequently, unknown unlabeled samples may receive erroneous known-class pseudo-labels during training, and unknown test samples are forced into one of the known classes. (c) Under the open-set setting, known samples are assigned to their corresponding classes, while samples outside the known-class acceptance regions are rejected as unknown.

made in graph learning-based methods [25], [26], generative model-based methods [27], [28], and pseudo-label generation methods [29], [30], [31]. Meanwhile, with the rapid development of large-scale representation learning, hyperspectral large models and advanced deep architectures have recently attracted increasing attention. Representative works, such as HyperSIGMA [56], MMD-MLP [57], SelfMTL [58], MAGS [59], MMFormer [60], TSCCD [61], and PHDMamba [62], have enhanced hyperspectral representation learning through large-scale pretraining, multimodal feature fusion, self-supervised or transfer learning, partial-label modeling, and long-range spectral–spatial dependency modeling. These methods demonstrate the strong potential of powerful feature representation in HSI classification and related interpretation tasks. However, most existing hyperspectral classification methods are still built upon the closed-set assumption; that is, they assume that unlabeled samples in both the training and testing stages come from the known class space and do not contain unknown class unseen during training [30], [32], [33], [34]. Fig. 1 illustrates the open-set semi-supervised HSI classification setting and its distinction from the conventional closed-set assumption. This assumption is often difficult to satisfy in real remote sensing scenarios because land-cover categories are continuously changing and complex, and limited labeled samples usually cannot cover all classes in the entire image. When unknown classes are mixed into unlabeled samples, traditional closed-set learning strategies tend to misclassify them into known classes, thereby contaminating the model's discriminative boundary, reducing known-class classification accuracy, and weakening the ability to reject unknown classes. To address this issue, this article introduces open-set semi-supervised HSI classification, aiming to train a model using unlabeled samples containing unknown class. The model is required to accurately predict known class while rejecting unknown class [35], [36].

Furthermore, the difficulty of open-set hyperspectral classification lies not only in the existence of unknown classes, but also in its close coupling with the inherent spectral–spatial characteristics of HSIs. On the one hand, HSIs contain rich structural cues. Land-cover objects, such as roads, rivers, field boundaries, and building contours, often exhibit significant directionality and complex boundaries. If traditional positional modeling methods are still adopted [27], [28], the expression of critical spatial priors may be weakened. On the other hand, due to imaging background, illumination, temperature, and other factors, different classes may have similar spectra [37], [38], while samples within the same class may also exhibit large spectral differences. As a result, unknown-class samples and boundary samples show strong uncertainty in prediction. Although existing open-set hyperspectral classification studies have explored distance-based methods [39], [40], [41], [42], reconstruction-based methods [43], [44], [50], and uncertainty-based methods [45], [46], most methods still mainly rely on fully supervised training or adopt fixed thresholds for sample screening [53], making it difficult to simultaneously consider complex spatial boundary modeling and effective utilization of limited labeled samples under the interference of unknown classes.

To address the problem that semi-supervised HSI classification cannot adapt to unknown classes, this study introduces the task of open-set semi-supervised HSI classification. Considering the challenges of complex spatial boundaries, interclass spectral similarity, and significant intraclass variability in HSIs, this article proposes an adaptive boundary consistency learning (ABCL) model. The proposed framework achieves low-cost and high-accuracy perception while simultaneously considering known-class classification and unknown-class rejection. First, to more fully describe directional spatial structures in local neighborhoods of HSIs, an anisotropic position transformer (APT) is designed. By decoupling horizontal and vertical positional biases, APT enhances the

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model's ability to represent complex land-cover boundaries and direction-sensitive spatial relationships. Second, to reduce the interference of unknown-class samples on consistency learning, an adaptive entropy mask (AEM) is proposed. It uses predictive entropy to characterize the uncertainty of unlabeled samples and combines batch-level statistical information to adaptively fuse hard selection and soft weighting, thereby enabling more robust utilization of reliable unlabeled sample information. Finally, a category recognition module (CRM) is introduced to further improve the discriminative ability of the feature space by enhancing intraclass compactness and interclass separability, providing a more stable representation basis for open-set discrimination. The main contributions can be summarized as follows.

1) To address the problem that semi-supervised HSI classification cannot adapt to unknown classes, an ABCL model is proposed. It achieves low-cost and high-accuracy perception while simultaneously considering known-class classification and unknown-class rejection, thereby promoting low-cost hyperspectral remote sensing target perception with high adaptability to unknown classes.
2) To enhance the model's ability to represent complex land-cover boundaries and spatial structural differences, an APT is designed to introduce direction-sensitive spatial priors by decoupling horizontal and vertical positional biases.
3) To reduce the interference of unknown classes in a large number of unlabeled samples, an AEM is proposed to guide consistency learning, thereby improving the discriminability of model features.
4) To further optimize the feature space structure, a CRM is introduced. By enhancing intraclass compactness and interclass separability, it provides a more stable representation basis for open-set discrimination.

The remainder of this article is organized as follows. Section II presents the related work. Section III describes the proposed ABCL model, covering the model overview, APT, AEM, CRM, and optimization objective. Section IV presents the experiments and results. Section V concludes the study.

II. Related Works

A. Semi-Supervised HSI Classification

HSIs contain high-dimensional spectral information and complex spatial structures. How to fully exploit the spectral–spatial information in unlabeled data under limited labeled samples has long been an important research direction in semi-supervised HSI classification. Existing semi-supervised HSI classification methods for this problem can mainly be divided into graph learning-based methods [25], [26], generative model-based methods [27], [28], and pseudo-label generation-based methods [29], [30], [31].

Graph learning-based methods [25], [26] usually construct pixel-level or superpixel-level graph structures to characterize the topological relationships among samples, and employ graph neural networks to extract discriminative features. Sha et al. [25] encoded spectral features and spatial coordinates based on graph attention networks for semi-supervised HSI classification. Qin et al. [26] designed a spectral–spatial graph convolutional network for semi-supervised learning on graph-structured data. However, these methods can only classify unlabeled samples involved in the model learning process, resulting in poor flexibility and generalization ability, which limits their practical applications.

Generative model-based methods [27], [28] usually employ generative adversarial networks (GANs) to learn the latent data distribution of unlabeled samples. Zhan et al. [27] proposed a GAN-based semi-supervised HSI classification method, but this method mainly relies on spectral features and fails to fully exploit spatial information. Zhong et al. [28] further combined GANs with conditional random fields (CRFs) for semi-supervised HSI classification. Although such methods have certain advantages in utilizing unlabeled data, mode collapse is a common problem during GAN training.

Pseudo-label learning methods [29], [30], [31] utilize high-confidence unlabeled samples to generate pseudo-labels for assisting model training. Kang et al. [29] trained a traditional classifier using a small number of labeled samples and selected high-confidence unlabeled samples as pseudo-labeled samples. Yao et al. [30] proposed a new contrastive learning strategy, in which low-confidence unlabeled samples were used as candidate negative samples of the corresponding classes. Xu et al. [31] constructed a self-ensembling framework composed of an ensemble network and a basic network, where pseudo-labels generated by the ensemble network were used to update the parameters of the basic network.

Nevertheless, most existing semi-supervised hyperspectral classification methods are still built upon the closed-set assumption, namely, they assume that unlabeled samples in both the training and testing stages come from the known class space and do not contain unknown classes unseen during training [30], [32], [33], [34]. However, in real remote sensing scenarios, land-cover categories are continuously changing and complex, and limited labeled samples often cannot cover all classes in the entire image. Unknown classes are therefore inevitably mixed into unlabeled samples. In this case, if all unlabeled samples are still treated as known classes for utilization, unknown-class samples may easily contaminate the decision boundary learned by the model, thereby reducing known-class classification accuracy and weakening the ability to reject unknown classes. Therefore, closed-set semi-supervised learning frameworks are often difficult to adapt to open environments.

B. Hyperspectral Large Models and Advanced Representation Learning

In recent years, HSI interpretation has been promoted by large-scale pretraining, multimodal fusion, self-supervised learning, and sequence modeling architectures. HyperSIGMA [56] is a representative hyperspectral foundation model that explores large-scale pretraining and unified representation learning for diverse HSI interpretation tasks, showing the potential of foundation models in hyperspectral remote sensing. In addition, several recent methods have further improved

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HSI classification from different perspectives. MMD-MLP [57] introduces a LiDAR-guided local–global directional MLP with multiresolution and multiscale representation to enhance HSI classification using complementary elevation information. MMFormer [60] designs a multimodal transformer with multiscale self-attention to fuse HSI and LiDAR features for remote sensing image classification. MAGS [59] focuses on partial label learning and alleviates label ambiguity in HSI classification through a max-gap loss-guided Siamese-reconstruction network. PHDMamba [62] introduces a progressive hybrid Mamba architecture to capture long-range spectral–spatial dependencies and adaptively fuse spectral and spatial information for HSI classification. Beyond classification, SelfMTL [58] and TSCCD [61] also demonstrate the effectiveness of self-supervised meta-transfer learning and temporal cross-domain learning in hyperspectral target detection and change detection, respectively, providing useful insights into transferable representation learning in hyperspectral analysis.

Although these methods have significantly advanced hyperspectral representation learning, most of them are mainly designed for closed-set classification, multimodal classification, target detection, or change detection scenarios. They usually assume that the testing categories are known in advance or focus on improving feature representation under supervised or task-specific settings. However, in practical open-set semi-supervised HSI classification, unlabeled samples may contain unknown categories, and directly using unreliable unlabeled samples may shift the decision boundary. Therefore, how to effectively exploit unlabeled data while suppressing unknown-class interference remains an important problem. Different from the above methods, the proposed ABCL focuses on open-set semi-supervised HSI classification by jointly modeling the physical spatial boundary and the feature decision boundary, and by using adaptive entropy-based uncertainty modeling to regulate the contribution of unlabeled samples.

C. Open-Set HSI Classification

In recent years, several HSI classification methods for open environments have been proposed. These methods mainly include distance-based methods [39], [40], reconstruction-based methods [43], [44], [50], and uncertainty-based methods [45], [46], which effectively extract the spectral–spatial features of HSIs.

Distance-based methods [39], [40], [41], [42] usually construct prototypes, class centers, or anchors for known classes in the feature space, and identify unknown classes using the distance or similarity between test samples and these representative vectors. Du et al. [39] introduced a feature invariance constraint into a prototype network and employed weighted Pearson distance for open-set recognition. Feng et al. [40] further constrained known-class features into a hyperspectral prototype space and combined prototype radii for unknown-class recognition. Xu et al. [41] used class anchors and orientational clustering learning to compact intraclass distributions, and then achieved known-class classification and unknown-class rejection based on the distance from samples to class centers. However, this type of method still depends heavily on the quality of prototype construction, the distance metric, and the rationality of threshold or boundary settings.

Reconstruction-based methods [43], [44], [50] mainly learn the reconstruction process of known-class samples and determine whether a test sample belongs to an unknown class according to the error or difference between the input and the reconstruction result. Yue et al. [43] reconstructed latent representations from spectral and spatial branches to preserve spectral–spatial details useful for unknown-class detection. Ma et al. [44] used a reconstruction module trained on the source domain to calculate the reconstruction discrepancy of target samples and converted it into similarity weights to assist unknown-class recognition and feature alignment in open-set domain adaptation. Liu et al. [50] designed a multitask learning framework to simultaneously perform classification and reconstruction, and combined extreme value theory to model reconstruction errors for distinguishing known classes from unknown classes. Such methods can, to some extent, overcome the limitation of relying only on classifier outputs for open-set discrimination, but they depend excessively on reconstruction quality.

Uncertainty-based methods [45], [46] usually distinguish known classes from unknown classes by estimating the uncertainty of model outputs. Their core idea is to utilize the fact that unknown samples often have higher predictive uncertainty to assist open-set discrimination. Ji et al. [45] introduced evidence theory into open-set hyperspectral classification, collected multiclass evidence through an improved GAN, calculated predictive uncertainty according to evidence strength, and then combined closed-set classification results to complete unknown-class discrimination. Xie et al. [46] simultaneously modeled feature similarity and sample uncertainty within a prototype learning framework, and integrated known-class probability and unknown-class probability through a probabilistic fusion strategy to improve the distinction between known and unknown classes.

Existing open-set HSI classification methods can extract certain spectral–spatial features and achieve high classification accuracy when sufficient labeled samples are available. However, under the condition of limited labeled samples, these methods still face two limitations. First, supervision information during model training is limited, which easily leads to overfitting and makes it difficult to learn stable open-set discriminative boundaries. Second, HSIs themselves contain complex spatial boundaries, interclass spectral similarity, and significant intraclass variability, making spectral–spatial feature representation and category discrimination more difficult. To address these problems, this study proposes an ABCL model to improve the overall performance of open-set semi-supervised HSI classification.

III. Methodology

A. Description of Problem

We define an HSI as $X \in \mathbb{R}^{H\times W\times D}$, where H, W, and D denote the height, width, and number of spectral bands of the image, respectively. The weight parameters of ABCL are trained using the labeled sample set $\mathcal{D}^L = \{(x_i^l, y_i^l)\}_{i=1}^{N^L}$ and

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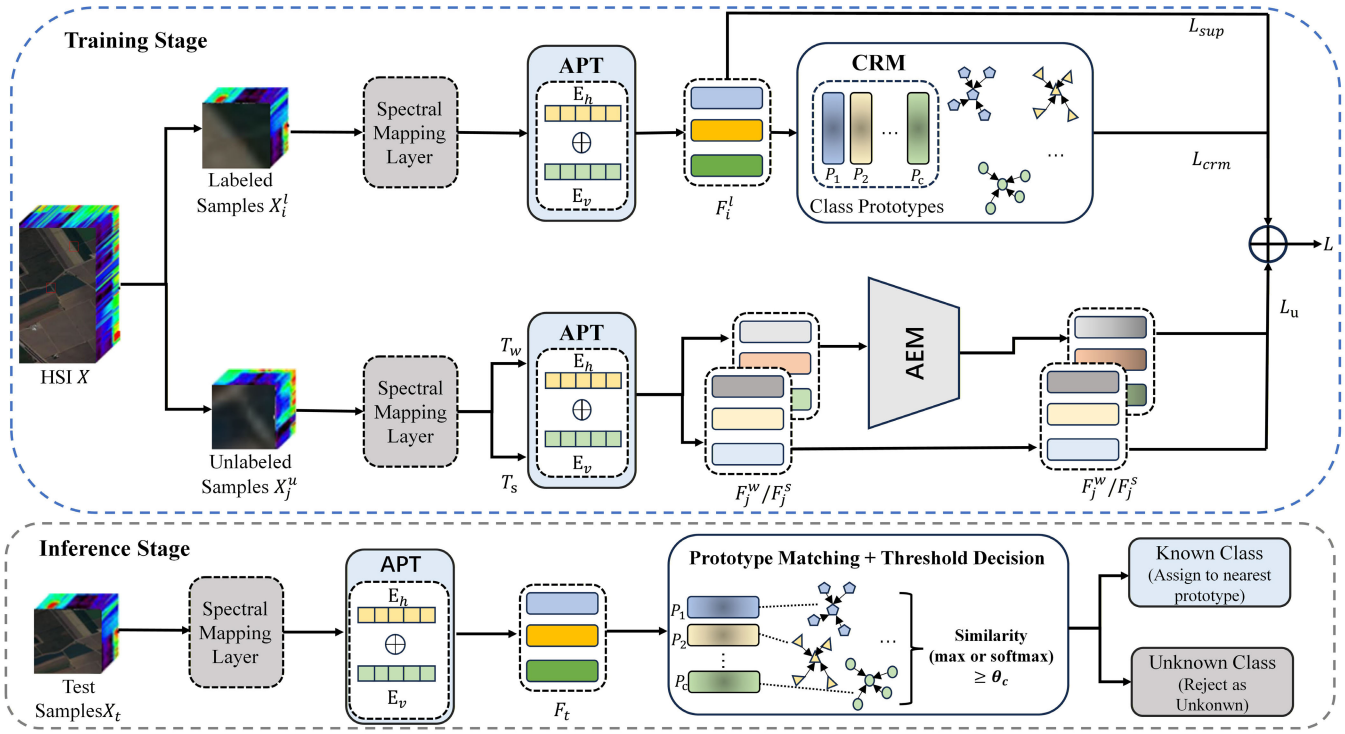


Fig. 2. Overall architecture of the proposed ABCL, including the training and inference stages. During training, labeled samples (X_i^l) and weakly and strongly augmented views of unlabeled samples (X_j^u) are jointly processed by the shared SML and APT. CRM constructs known-class prototypes (P_c) from labeled features (F_i^l), while AEM weights the consistency learning between weak-view and strong-view features (F_j^w) and (F_j^s). During inference, test samples (X_t) are processed by the trained SML and APT to obtain its feature (F_t). Its prediction is accepted when the distance between (F_t) and the corresponding class prototype does not exceed the class-specific threshold (θ_c); otherwise, (X_t) is rejected as unknown. (T_w) and (T_s) denote the weak and strong transformations; (E_h) and (E_v) denote the learnable horizontal and vertical positional parameter matrices; and ($\mathcal{L}$sup), ($\mathcal{L}$crm), ($\mathcal{L}_u$), and ($\mathcal{L}$) denote the supervised, prototype-based, consistency, and total losses, respectively.

the unlabeled sample set $\mathcal{D}^U = \{(x_j^u)\}_{j=1}^{N^U}$, where $x_i^l/x_j^u \in \mathbb{R}^D$ denotes the ith/jth labeled/unlabeled pixel, $y_i^l \in \mathbb{R}^C$ denotes the land-cover category of x_i^l, and C represents the number of known classes. Moreover, N^L and N^U denote the numbers of labeled and unlabeled samples, respectively, with $N^L \ll N^U$.

Different from traditional closed-set semi-supervised classification, this article studies the problem of open-set semi-supervised HSI classification. That is, unlabeled samples $\mathcal{D}^U$ are no longer assumed to share exactly the same category space as labeled samples $\mathcal{D}^L$, and they may contain unknown-class samples that do not appear in the labeled samples during training. In other words, labeled samples only cover known classes, whereas unlabeled samples may come from both known and unknown classes. Therefore, the model needs not only to learn discriminative features of known classes by utilizing a small number of labeled samples and a large number of unlabeled samples, but also to effectively reject samples that deviate from the known-class distribution during testing, thereby simultaneously accomplishing known-class recognition and unknown-class detection.

B. Overview of the Proposed ABCL

The proposed ABCL mainly consists of the APT, the AEM, and the CRM. Fig. 2 shows the overall architecture of the proposed ABCL model, including both the training stage and the inference stage. This model aims to simultaneously optimize the physical spatial boundary and the feature decision boundary under the semi-supervised open-set scenario. During training, ABCL contains two input branches with different roles. The upper branch takes labeled HSI samples as input and is mainly used for supervised feature learning and class prototype construction. Since HSI data are highly redundant, a spectral mapping layer (SML) is first adopted for preliminary dimensionality reduction. Then, the dimension-reduced features are fed into APT to capture spatial structural information. APT decouples the spatial relative positional bias into horizontal and vertical components, effectively sharpening the physical spatial boundaries of complex land-cover objects. The extracted high-level labeled features are further projected into a latent metric space, where CRM constructs class prototypes for each known class. By pulling features of the same class closer and pushing features of different classes farther apart, CRM improves intraclass compactness and interclass separability.

The lower branch takes unlabeled HSI samples as input and is used for consistency-based semi-supervised learning. Specifically, weak and strong transformations are applied to each unlabeled sample, and the two augmented views are fed into the shared SML and APT to obtain corresponding feature representations. AEM is then designed to adaptively modulate the contribution of unlabeled samples according to prediction uncertainty, so that reliable unlabeled samples contribute more to the consistency learning process. It should be noted that the

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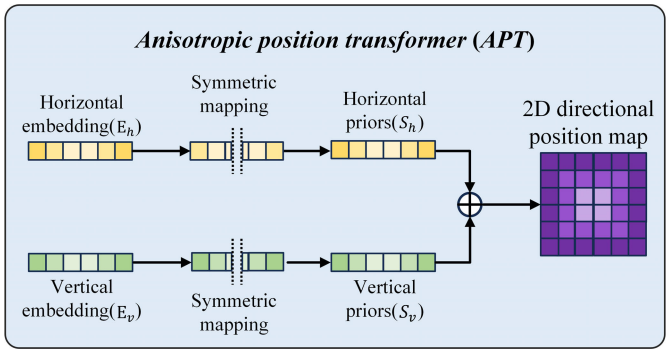


Fig. 3. Detailed diagram of the APT.

labeled and unlabeled branches are jointly optimized during training rather than being executed as two separate stages. The upper branch provides supervised prototype guidance, while the lower branch exploits unlabeled samples to improve the robustness and discriminability of the learned features.

During inference, the process is different from the training stage. The unlabeled branch, AEM-based consistency learning, and loss computation are no longer used. Given a test sample, ABCL first extracts its feature representation through the trained SML and APT. Then, a closed-set prediction is obtained, and the distance between the test feature and the prototype of the predicted class is compared with the corresponding class-specific threshold. If the distance is within the threshold, the sample is assigned to the predicted known class; otherwise, it is rejected as an unknown sample. In this way, ABCL uses the learned class prototypes and adaptive thresholds to achieve both known-class classification and unknown-class rejection.

C. Anisotropic Position Transformer

To more fully characterize the directional spatial structures in local neighborhoods of HSIs and enhance the model's ability to represent complex land-cover boundaries, this article designs an APT. Hyperspectral classification usually takes a neighborhood patch centered on a target pixel as input, and neighboring pixels can provide important spectral–spatial contextual information for the center pixel. However, land-cover objects, such as roads, rivers, field boundaries, and building contours, often exhibit different spatial extension patterns in different directions. If traditional positional modeling methods [27], [28] are still adopted, it is difficult to fully express such direction-sensitive structural differences. Based on the high redundancy of HSI data, the SML first uses 1×1 convolutional blocks, the GeLU activation function, and batch normalization to perform preliminary dimensionality reduction to obtain a shallow spectral-space feature map. Then, as shown in Fig. 3, APT explicitly introduces direction-sensitive spatial priors into the shallow spectral-space feature map by decoupling the position modeling methods in the horizontal and vertical directions, thereby improving the model's ability to model complex boundaries and local structural differences.

There exist spatial positional priors worth modeling between the center pixel and neighboring pixels in HSIs. In particular, positions closer to the center pixel are usually more important, while pixels located at spatially symmetric positions should have similar positional relationships. Therefore, we transform the original learnable vectors through a specially designed symmetric mapping function to construct horizontal and vertical bias vectors

$$S_h = \phi(E_h), \quad S_v = \phi(E_v) \tag{1}$$

where E_h, E_v denote the original learnable positional parameter matrices initialized by the model in the horizontal direction, namely the column direction, and the vertical direction, namely the row direction, respectively. $\phi(\cdot)$ denotes the symmetric parameterization operator. S_h, S_v denote the anisotropic spatial bias vectors after symmetrization.

To capture the directional structural features of HSIs in the horizontal and vertical directions, APT fuses the decoupled horizontal and vertical components through tensor addition, generating a 2-D distance matrix capable of representing asymmetric spatial distributions

$$D_{k,i,j} = S_{v,k,i} + S_{h,k,j} \tag{2}$$

where D denotes the finally generated anisotropic spatial distance matrix with the dimension of $[d, H, W]$; k denotes the feature channel index; and $k \in \{1, \ldots, d\}$; $S_{v,k,i}$, $S_{h,k,j}$ represent the vertical spatial contribution of the ith row and the horizontal spatial contribution of the jth column under the kth feature dimension, respectively.

To enable the network to dynamically adjust the feature aggregation process according to the learned spatial priors, we directly inject the generated anisotropic distance matrix into the self-attention calculation of the transformer as a bias term. This mechanism enables the model to explicitly consider the geometric relative positions of pixels when computing their similarities, thereby effectively sharpening category boundaries and suppressing over-smoothing

$$A_{i,j} = \frac{Q_i K_j^T}{\sqrt{d_k}} + \text{Reshape}(D)_{i,j} \tag{3}$$

where $A_{i,j}$ denotes the reconstructed anisotropic attention score, which determines the weight allocation ratio of the center pixel to pixels in different directions when aggregating neighborhood information; Q_i, K_j represent the query vector and key vector in the self-attention mechanism; d_k denotes the dimension of the feature vector; and $Reshape(D)_{i,j}$ denotes the operation of reshaping the multidimensional distance matrix and mapping it into the attention map space. Through the above design, APT learns horizontal and vertical positional features separately and combines them into anisotropic spatial positional encoding, enabling the model to perceive neighborhood relationships in different directions more finely, thereby more accurately characterizing complex spatial boundaries and structural information in HSIs.

D. Adaptive Entropy Mask

In open-set semi-supervised hyperspectral classification, unlabeled samples inevitably contain unknown classes. If they are directly used in consistency learning, the model decision

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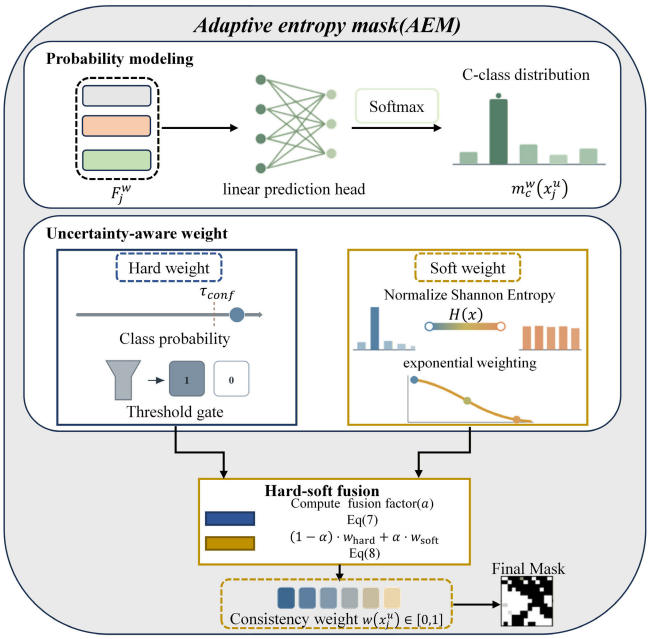


Fig. 4. Illustration of the AEM. Softmax categorical probabilities are converted into confidence-based hard weights and entropy-based soft weights, which are adaptively fused to generate the final consistency weight.

boundary may be shifted, thereby reducing classification performance. Existing methods usually perform hard selection based on a fixed confidence threshold and only use high-confidence samples for training [53]. However, this strategy ignores the information of samples near the decision boundary, reducing the efficiency of unlabeled data utilization and failing to adapt to data distribution differences across various scenarios. Therefore, from the perspective of decision-boundary uncertainty modeling, this article proposes an AEM, as shown in Fig. 4. This module measures sample uncertainty based on the normalized information entropy of the prediction distribution of unlabeled samples. Samples with stable predictions and high confidence are assigned larger training weights, while samples with high uncertainty, which may belong to unknown classes or lie near category boundaries, are assigned lower participation strength in the consistency constraint.

Specifically, normalized Shannon entropy is used as a metric to measure model prediction uncertainty. This metric uses the dispersion degree of the predicted probability distribution to characterize the discriminative stability of the model for a sample. When the predicted probability is concentrated on a certain class, the model is highly certain. When the probability distribution is relatively flat, the sample is more likely to be near the category boundary or to be affected by unknown-class interference, thus exhibiting higher uncertainty. Compared with a single maximum probability, normalized Shannon entropy can characterize the confusion degree of prediction results from the overall distribution level. Given the weakly augmented feature F_j^w produced by APT, AEM first uses the linear prediction head to generate class logits and converts them through Softmax into a categorical predictive distribution over the C known classes. Let $g_c(\cdot)$ denote the cth output of the linear prediction head. The categorical probability that the jth unlabeled sample belongs to the cth known class and its corresponding normalized Shannon entropy are defined as

$$m_c^w\left(x_j^u\right) = \frac{\exp\left(g_c\left(F_j^w\right)\right)}{\sum_{k=1}^{C}\exp\left(g_k\left(F_j^w\right)\right)}, \qquad c = 1,\ldots,C \tag{4a}$$

$$H\left(x_j^u\right) = -\frac{1}{\log C}\sum_{c=1}^{C} m_c^w\left(x_j^u\right)\log\left(m_c^w\left(x_j^u\right)\right) \tag{4b}$$

where $\mathbf{m}^w(x_j^u) = [m_1^w(x_j^u),\ldots,m_C^w(x_j^u)]$ denotes the weak-view categorical predictive distribution, and $\log C$ denotes the maximum possible information entropy. Because the land-cover category is a discrete variable, its predictive probability is modeled as a categorical distribution. Therefore, the entropy is calculated directly from the Softmax probabilities rather than from an additionally estimated continuous probability density function.

A larger $H(x)$ indicates that the model is more uncertain about the category assignment of this pixel, and the probability that it belongs to an unknown class is extremely high. Based on this, we construct a nonlinear exponential soft weight for each sample

$$w_{\text{soft}}\left(x_j^u\right) = \exp\left(-\lambda_s H\left(x_j^u\right)\right) \tag{5}$$

where λ_s is a hyperparameter that controls the weight decay rate.

The inherent signal-to-noise ratio of different hyperspectral datasets varies greatly, and fixed hyperparameters cannot achieve cross-scenario generalization. Therefore, in each iteration, the AEM module dynamically calculates the average normalized entropy of unlabeled samples in the current batch and determines the overall uncertainty degree of the batch accordingly

$$\tilde{E} = \frac{1}{N^U}\sum_{j=1}^{N^U} H\left(x_j^u\right) \tag{6}$$

where N^U denotes the total number of unlabeled samples in the current batch. Based on the overall uncertainty of the current batch, we design a dynamic fusion factor α to adaptively adjust the proportion between hard truncation and soft weighting

$$\alpha = clamp\left(\frac{\tilde{E}-\tau_1}{\tau_2-\tau_1}, 0, 1\right) \tag{7}$$

where $clamp(\cdot)$ denotes interval clipping, and τ_1, τ_2 denote the entropy thresholds for clean and chaotic environments, respectively. When $\tilde{E} \le \tau_1$ holds, α approaches 0, and the model tends toward conservative hard truncation. When $\tilde{E} \ge \tau_2$ holds, α approaches 1, and the model adaptively activates the soft mask mechanism.

To balance training stability and boundary modeling ability, a strict hard selection mechanism is adopted in the early stage of training to ensure prototype purity, while soft weights are gradually introduced in the later stage to fully utilize boundary sample information. Finally, the adaptive weight is jointly determined by the hard mask and the soft weight

$$w\left(x_j^u\right) = (1-\alpha)\cdot w_{\text{hard}}\left(x_j^u\right) + \alpha\cdot w_{\text{soft}}\left(x_j^u\right) \tag{8}$$

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where $w_{hard}(x_j^u) = \mathbb{I}(m_c^w(x_j^u) > \tau_{conf})$ denotes the traditional confidence-based hard mask; $\mathbb{I}(\cdot)$ denotes the indicator function; τ_{conf} represents a fixed confidence level.

In summary, AEM realizes uncertainty-driven adaptive weight allocation and dynamic modeling of the decision boundary. It enables the model to fully utilize unlabeled data to guide consistency learning while effectively suppressing the interference of unknown classes, thereby improving open-set classification performance. Its consistency loss is formulated as

$$\mathcal{L}_{\mathrm{u}} = \frac{1}{N^U} \sum_{j=1}^{N^U} w\left(x_j^u\right) \cdot \mathrm{CE}\left(F_j^s, F_j^w\right) \tag{9}$$

where F_j^s, F_j^w denote the prediction results of unlabeled samples under strong and weak augmentations, respectively, and $w(x_j^u)$ denotes the adaptive weight generated by the AEM module. AEM dynamically regulates the consistency learning process using sample uncertainty. Specifically, it enhances the contribution of high-confidence samples to parameter updates and reduces the disturbance of high-uncertainty samples to the model discriminative boundary, thereby improving the reliability and effectiveness of unlabeled sample utilization in open-set semi-supervised scenarios.

E. Category Recognition Module

Due to the diversity of land-cover categories, HSIs often have both complex spatial structures and significant spectral variations. In addition, affected by imaging conditions, land-cover mixing, background interference, and other factors, HSI data commonly exhibit large intraclass differences and interclass spectral similarity. Such complex spectral–spatial distributions easily lead the model to form loose or even overlapping category distributions in the feature space, making known and unknown classes more likely to be confused in boundary regions. Existing studies [34], [47] have shown that samples from the same class usually tend to cluster in the feature space, while samples from different classes are relatively dispersed. Therefore, class prototypes can be used to model the central representations of known classes, and prototype constraints can be further employed to enhance intraclass compactness and interclass separability. Based on this idea, this article introduces the CRM, which constrains the feature space based on the construction of class prototypes, thereby improving the discriminative ability of the model in open-set scenarios.

For the cth known class, let $\mathcal{B}_c$ denote the sample index set of this class in the current batch. Then, the prototype of this class is defined as the mean of the corresponding labeled sample features

$$P_c = \frac{1}{|\mathcal{B}_c|} \sum_{i \in \mathcal{B}_c} F_i^l \tag{10}$$

where P_c denotes the prototype of the cth class, and F_i^l denotes the labeled feature.

After obtaining the prototypes of all classes, this article uses the Manhattan distance between sample features and each class prototype to measure similarity. To smoothly transform the distance metric into an optimizable probability distribution form, this article adopts a similarity mapping based on the Gaussian kernel form

$$s\left(F_i^l, P_c\right) = \exp\left(-\frac{\mathrm{Dis}\left(F_i^l, P_c\right)^2}{T}\right) \tag{11}$$

where $Dis(\cdot)$ denotes the Manhattan distance function, and T denotes the temperature coefficient.

If the deviation degree between an unlabeled sample D^U and all known-class prototypes P_c exceeds the threshold θ_c, it is labeled as an unknown class; otherwise, it is labeled as a known class. For the cth known class, the Manhattan distance function is used to calculate the deviation degree between labeled sample features and the corresponding class prototype, which is then fit into a Gaussian distribution s. Then, labeled samples of other classes are regarded as negative samples of the cth class, and the deviation degrees between these negative samples and the prototype of the cth class are fit into another Gaussian distribution s^-. Finally, the value corresponding to the intersection point of s and s^- is used as the segmentation threshold θ_c of the cth class [54].

On this basis, the cross-entropy loss is adopted for class supervision

$$\mathcal{L}_{\mathrm{crm}} = -\frac{1}{N^L} \sum_{i=1}^{N^L} \sum_{c=1}^{C} \mathbb{I}\left(y_i^l = c\right) \log \frac{s\left(F_i^l, P_c\right)}{\sum_{m=1}^{C} s\left(F_i^l, P_m\right)} \tag{12}$$

where N^L and C denote the number of labeled samples and known classes, respectively; P_m denotes the prototype of the mth known class; and $\mathbb{I}(y_i^l = c)$ is an indicator function that equals 1 when the true class of the ith labeled sample is c, and 0 otherwise. For each labeled feature F_i^l, the ratio in (12) normalizes its similarity to P_c by the sum of its similarities to all known-class prototypes, thereby forming a prototype-based class distribution. The indicator function retains only the term corresponding to the ground-truth class. Therefore, (12) calculates the average cross-entropy between the ground-truth labels and the prototype-based class distribution. Minimizing $\mathcal{L}_{\mathrm{crm}}$ increases the relative similarity between a labeled feature and its corresponding class prototype while reducing its relative similarity to the other prototypes, thereby promoting intraclass compactness and interclass separability.

The total loss function of the proposed ABCL optimization objective is formulated as

$$\mathcal{L} = \mathcal{L}_{\mathrm{sup}} + \mathcal{L}_u + \mathcal{L}_{\mathrm{crm}} \tag{13}$$

where $\mathcal{L}_{\mathrm{sup}}$, $\mathcal{L}_{\mathrm{u}}$, and $\mathcal{L}_{\mathrm{crm}}$ denote the supervised loss, the unlabeled-sample consistency loss, and the category relationship constraint loss, respectively.

IV. Experiment and Results

A. Dataset and Evaluation Metrics

To verify the effectiveness of the proposed method in the open-set semi-supervised HSI classification task, experiments are conducted on three public hyperspectral datasets: Indian Pines (IP), Salinas (SA), and LN02. Regarding the open-set configuration, classes 3 and 6 in the IP dataset are designated as unknown classes, while the remaining categories are treated

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TABLE I
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE IP DATASET

Class	10 Training Samples Per Class							20 Training Samples Per Class						
	SpectralFormer	SSFTT	MDL4OW	SSMLP-RPL	SGMAEs	DSCA-Net	ABCL	SpectralFormer	SSFTT	MDL4OW	SSMLP-RPL	SGMAEs	DSCA-Net	ABCL
1	80.33	99.45	95.13	95.83	**100.00**	80.38	96.39	92.62	98.46	99.23	98.07	**100.00**	96.67	99.62
2	31.16	66.30	57.02	61.16	69.81	63.48	**72.74**	46.74	71.56	77.54	66.02	81.75	64.53	**87.23**
4	55.56	94.88	83.61	92.42	**94.97**	86.15	87.13	80.92	92.07	84.15	92.76	**99.31**	91.65	96.28
5	51.58	86.13	80.54	81.56	83.29	88.59	**97.31**	71.17	90.13	90.71	86.52	87.74	93.36	**96.31**
7	98.02	**100.00**	**100.00**	98.88	83.86	72.46	83.26	99.45	**100.00**	99.45	95.56	92.37	74.08	93.09
8	87.70	97.03	92.71	95.02	94.67	**98.67**	92.89	94.18	**99.06**	98.25	93.18	96.31	98.49	97.67
9	**100.00**	**100.00**	**100.00**	98.01	99.45	91.11	99.44	**100.00**	**100.00**	**100.00**	99.00	**100.00**	98.33	**100.00**
10	45.73	76.57	80.76	76.71	83.42	92.45	**99.02**	73.12	84.39	87.78	76.15	89.61	96.13	**99.02**
11	54.27	70.69	64.66	67.43	71.83	72.00	**98.00**	51.27	76.87	70.97	68.95	73.68	95.87	**100.00**
12	35.14	70.46	68.01	65.61	79.18	58.80	**80.69**	51.39	78.32	84.76	75.37	87.99	68.35	**90.05**
13	93.88	**99.53**	87.74	98.67	84.53	62.89	82.00	97.42	**99.35**	94.21	97.78	85.22	72.92	87.21
14	54.86	**89.18**	81.33	80.02	75.16	66.35	81.23	71.52	91.42	90.85	86.01	87.67	74.14	**92.84**
15	34.47	89.52	88.27	87.45	89.33	**99.49**	97.79	60.67	89.48	92.21	89.69	93.62	97.73	**99.08**
16	93.27	98.43	90.12	97.71	88.71	**98.67**	88.57	**98.20**	89.86	83.15	97.53	95.59	94.84	95.50
Unknown	4.75	7.11	5.72	24.32	29.28	4.50	**58.24**	3.38	6.69	4.53	47.26	24.21	4.62	**55.07**
OA	44.56±2.38	67.26±3.51	61.73±3.21	66.53±1.74	72.89±2.77	62.14±2.46	**79.56±2.79**	56.61±2.18	70.83±3.89	70.29±0.86	72.43±1.93	78.64±0.98	64.88±0.86	**85.53±1.38**
AA	61.38±3.20	83.02±2.67	78.38±1.79	81.39±1.31	81.83±1.55	75.73±1.95	**87.65±1.82**	72.80±1.57	84.51±3.38	83.85±1.34	84.66±1.57	86.34±0.75	81.45±1.91	**92.60±0.78**
Kappa	36.39±2.45	62.55±3.95	53.82±3.38	61.89±2.02	69.16±3.08	57.54±2.23	**76.28±3.20**	52.02±2.64	66.55±4.40	66.39±2.18	68.33±2.27	75.63±1.19	60.17±2.94	**83.24±1.57**

as known classes; for the SA dataset, classes 4, 12, and 14 are set as unknown classes, with the rest considered known classes; for the LN02 dataset, class 8 is designated as the unknown class, while the remaining categories are regarded as known classes. During the training phase, only a small number of labeled samples are randomly extracted from the known classes to participate in supervised learning, while a large amount of unlabeled samples are retained to construct a semi-supervised open-set scenario. Specifically, this article sets the extraction of 10 and 20 hyperspectral pixels from each known class as labeled samples, respectively. In the IP dataset, due to the limited sample size of class 7 (grass-pasture-mowed) and class 9 (oats), a fixed number of 10 labeled samples is extracted for these two classes across all experiments. Such settings can more adequately verify the model's generalization ability and open-set robustness under conditions with extremely scarce labeled data. The evaluation metrics employ overall accuracy (OA), average accuracy (AA), and the Kappa coefficient, which are commonly used in open-set environments, to comprehensively evaluate the model's performance in recognizing known classes and rejecting unknown classes.

B. Experimental Settings

The experiments in this study are implemented based on the PyTorch framework and trained using two NVIDIA GeForce RTX 2080Ti GPUs. The feature dimension of ABCL is set to 64, and the number of heads in the multihead self-attention mechanism is set to 4. The entropy thresholds τ_1 and τ_2 are set to 0.1 and 0.3, respectively. The ratio of unlabeled samples to labeled samples is set to 4, and the batch size is set to 128. The number of training iterations is set to 10 000, which corresponds to 100 epochs, with each epoch comprising 100 parameter updates. In the first 50 epochs, a strict hard screening mechanism (with the confidence threshold τ_{conf} set to 0.95) is adopted to ensure the purity of the prototypes, while in the subsequent 50 epochs, soft weights are introduced to fully utilize the information of boundary samples. The Adam optimizer is employed, and the learning rate is fixed at 0.0005. All experiments are repeated ten times, and the average value of the ten independent experimental results is taken as the final performance to mitigate the fluctuations caused by random sampling.

C. Comparative Experimental Results

To ensure a comprehensive comparison, this article simultaneously selects native open-set methods and representative closed-set methods as comparison baselines, including SpectralFormer [48], SSFTT [49], MDL4OW [50], SSMLP-RPL [51], SGMAEs [52], and DSCA-Net [55]. For a fair comparison, the experimental settings for MDL4OW and SSMLP-RPL follow their original articles, whereas for the other methods, unlabeled samples with a confidence score greater than 0.5 are regarded as known classes; otherwise, they are treated as unknown classes.

1) Results Analysis on the IP Dataset: Extensive experiments were conducted on the IP dataset to validate the effectiveness of ABCL. Table I presents the classification results of different methods on the IP dataset. It can be observed that, under the setting with only ten labeled samples per known class, ABCL achieves 58.24%, 79.56%, 87.65%, and 76.28% in terms of unknown, OA, AA, and Kappa, respectively, significantly outperforming all compared methods. From the perspective of class-wise accuracy, ABCL achieves clear advantages on several challenging categories, substantially surpassing the competing methods. These categories usually exhibit more complex boundary distributions or stronger interclass similarity, making them more susceptible to the interference of unknown-class samples and boundary noise. In contrast, closed-set transformer-based methods, such as SpectralFormer and SSFTT, can extract certain spectral–spatial features, but they assume that all samples belong to known classes, resulting in weak unknown-class recognition capability in open-set environments. Although MDL4OW and SSMLP-RPL possess open-set recognition capability, they still insufficiently exploit potential unknown-class samples in the unlabeled data. SGMAEs and DSCA-Net can utilize unlabeled information, but it remains vulnerable to unknown-class interference in open-set scenarios. By

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Fig. 5. Classification maps of different methods on the IP dataset. (a) Ground truth. (b) SpectralFormer. (c) SSFTT. (d) MDL4OW. (e) SSMLP-RPL. (f) SGMAEs. (g) DSCA-Net. (h) Proposed ABCL.

modeling direction-sensitive spatial priors through APT and adaptively regulating the contribution of unlabeled samples to consistency learning through AEM, ABCL substantially improves the model's discrimination ability for complex categories.

When the number of labeled samples per known class increases to 20, the overall performance of all methods improves, while ABCL still maintains the best results. Under this setting, ABCL achieves 55.07%, 85.53%, 92.60%, and 83.24% in terms of unknown, OA, AA, and Kappa, respectively. It is worth noting that as the number of labeled samples increases, the fitting capability of all methods for known classes is enhanced. Nevertheless, ABCL can still significantly improve unknown-class rejection while maintaining high known-class recognition accuracy, indicating that its adaptive boundary modeling mechanism has good generalization ability and stability. Overall, the IP dataset contains complex spatial boundaries and strong category mixing, making it highly challenging under open-set conditions with limited labeled samples. ABCL achieves larger performance gains over the baselines on this dataset, demonstrating its effectiveness in modeling complex spatial structures and boundary uncertainty.

Fig. 5 shows the classification maps of different methods on the IP dataset. Compared with the other methods, ABCL produces classification maps with better spatial consistency and boundary preservation on the IP dataset. Specifically, closed-set methods, such as SpectralFormer and SSFTT exhibit obvious scattered misclassification and region fragmentation in complex areas. Although MDL4OW, SSMLP-RPL, SGMAEs, and DSCA-Net improve structural integrity in some regions, a certain degree of category confusion can still be observed at land-cover intersections and in small local regions. In contrast, the classification results of ABCL are smoother and more coherent, with fewer noisy pixels within homogeneous regions, and the spatial contours around complex boundaries are better preserved. This indicates that ABCL can learn more stable discriminative representations under open-set semi-supervised conditions. As shown in the red dashed boxes in Fig. 5, the proposed ABCL can accurately identify known classes and unknown classes, and its classification map is smoother than those of the other methods. This phenomenon is consistent with the design of the proposed method: APT strengthens direction-sensitive spatial structure modeling, AEM alleviates the interference of unknown-class samples on consistency learning, and CRM further enhances intraclass compactness and interclass separability. Therefore, ABCL can simultaneously improve the stability of known-class classification and the discriminability of unknown classes in complex scenes.

2) Results Analysis on the SA Dataset: Table II presents the classification results of different methods on the SA dataset. Similar to the IP dataset, ABCL achieves the best overall performance when ten and 20 labeled samples are selected for each known class. Under the setting with only ten labeled samples per known class, ABCL obtains 37.35%, 88.20%, 91.14%, and 86.87% in terms of unknown, OA, AA, and Kappa, respectively. From the class-wise perspective, ABCL performs particularly well on several categories with relatively high classification difficulty, substantially outperforming all compared methods. This indicates that although some land-cover regions in the SA dataset show large-area homogeneous

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TABLE II
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE SA DATASET

Class	10 Training Samples Per Class							20 Training Samples Per Class						
	SpectralFormer	SSFTT	MDL4OW	SSMLP-RPL	SGMAEs	DSCA-Net	ABCL	SpectralFormer	SSFTT	MDL4OW	SSMLP-RPL	SGMAEs	DSCA-Net	ABCL
1	92.73	99.43	85.16	90.44	**99.82**	95.57	96.28	94.40	99.67	95.65	96.73	**99.94**	97.58	98.65
2	94.86	99.89	93.48	99.59	**99.95**	96.92	98.73	98.48	99.53	97.08	97.89	**99.99**	98.56	99.46
3	65.27	99.82	92.91	**100.00**	99.97	95.32	97.66	82.60	99.96	99.07	**100.00**	**100.00**	**100.00**	98.98
5	91.17	97.73	94.21	**98.76**	97.08	94.04	90.57	94.15	**99.74**	94.71	96.17	99.16	95.90	90.73
6	**99.52**	98.65	85.71	85.54	97.43	90.08	96.61	**99.91**	99.65	91.38	93.89	98.93	93.31	98.85
7	88.84	**99.56**	95.53	98.57	97.15	96.41	95.21	92.72	99.83	96.14	98.00	**99.96**	97.57	98.08
8	45.94	80.94	64.65	75.15	54.98	80.36	**97.13**	68.01	76.55	67.43	78.59	79.14	83.28	**98.89**
9	89.67	**99.88**	97.58	92.70	80.32	87.89	84.20	96.51	**99.93**	98.89	95.11	90.91	89.85	91.15
10	62.69	91.52	85.02	95.56	95.20	89.72	**96.08**	87.71	95.71	93.25	93.43	95.46	97.78	**99.01**
11	56.75	**99.40**	94.46	85.60	97.12	85.61	90.99	92.30	**99.88**	96.86	99.23	98.02	88.08	95.43
13	99.64	99.92	99.71	97.57	**100.00**	98.24	99.57	98.72	99.92	99.70	**100.00**	**100.00**	99.50	99.34
15	66.26	75.74	74.36	90.13	85.96	91.90	**96.77**	53.85	90.16	82.52	73.28	89.06	**98.62**	97.34
16	77.95	97.03	95.30	84.97	**99.25**	89.57	98.84	86.52	98.46	96.16	93.22	99.21	93.14	**99.53**
Unknown	9.72	3.96	3.74	25.11	5.47	1.80	**37.35**	5.96	2.68	15.69	32.82	3.12	2.46	**35.27**
OA	67.81±4.51	79.04±1.31	76.77±3.60	83.87±2.02	80.59±0.50	83.16±2.65	**88.20±1.49**	75.04±1.59	85.43±2.25	80.52±2.53	85.54±0.56	86.17±0.46	86.99±1.52	**90.55±1.47**
AA	74.36±3.68	88.82±0.59	82.99±2.03	87.12±1.03	86.41±0.20	85.25±1.21	**91.14±1.51**	82.27±0.92	90.12±0.78	87.46±1.01	89.17±0.46	89.49±0.17	88.26±0.58	**92.91±1.12**
Kappa	64.27±4.86	77.22±1.42	74.23±3.95	82.08±2.17	78.49±0.55	81.16±1.90	**86.87±1.65**	72.36±1.53	83.82±2.44	78.39±2.74	83.88±0.63	84.64±0.50	84.61±0.43	**88.58±1.63**

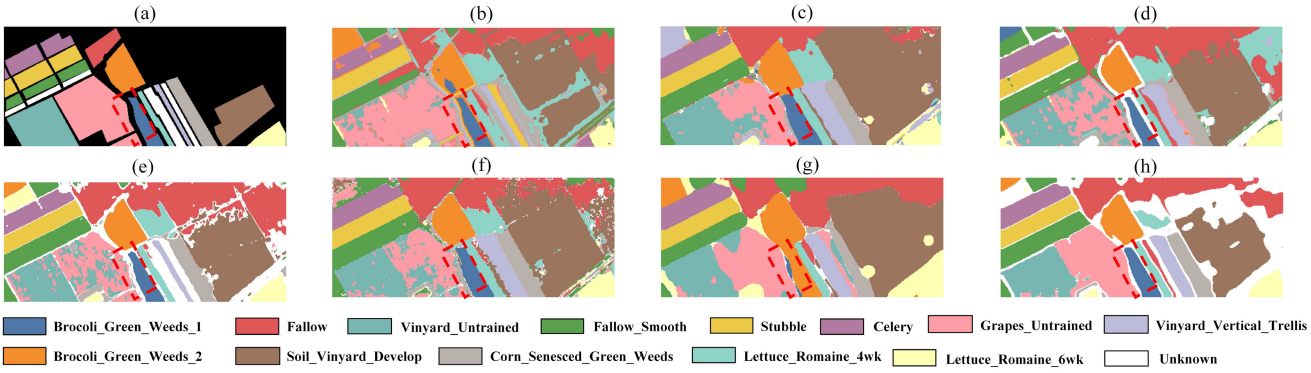


Fig. 6. Classification maps of different methods on the SA dataset. (a) Ground truth. (b) SpectralFormer. (c) SSFTT. (d) MDL4OW. (e) SSMLP-RPL. (f) SGMAEs. (g) DSCA-Net. (h) Proposed ABCL.

distributions, ABCL can still learn more discriminative feature representations for categories with blurred boundaries or high confusion under limited labeled samples. In contrast, closed-set methods still struggle to effectively handle unknown-class samples in open-set environments. Existing open-set methods can achieve unknown-class rejection to some extent, but their overall performance is still limited because they lack adaptive modeling of the uncertainty of unlabeled samples.

When the number of labeled samples per known class increases to 20, ABCL further achieves an unknown accuracy of 35.27%, an OA of 90.55%, an AA of 92.91%, and a Kappa of 88.58%. Meanwhile, ABCL still maintains clear advantages in challenging categories, indicating that as the number of labeled samples increases, the model can not only further improve its fitting capability for known classes but also maintain strong discriminability for unknown classes. Although the overall scene of the SA dataset is relatively regular and strong baseline methods can already achieve high accuracy, ABCL still remains superior in comprehensive metrics, further verifying the effectiveness of the proposed method. Overall, compared with the IP dataset, the SA dataset has higher overall separability, and therefore, the performance gaps among different methods are relatively smaller. Nevertheless, ABCL still achieves the best results in terms of OA, AA, Kappa, and unknown, demonstrating that the proposed method is not only suitable for scenes with complex boundaries and obvious category mixing but can also maintain stable advantages on relatively regular datasets. These results indicate that the synergy between APT and AEM can effectively enhance the feature representation capability and open-set discriminative ability of the model, while CRM further improves intraclass compactness and interclass separability, jointly contributing to the overall performance improvement.

Fig. 6 shows the classification maps of different methods on the SA dataset. Compared with the IP dataset, the SA dataset has a more regular scene structure, and thus the classification results of all methods are generally smoother. However, clear differences still exist under open-set semi-supervised conditions. As shown in the figure, SpectralFormer and SSFTT still produce many scattered misclassified pixels in some regions. Although MDL4OW, SSMLP-RPL, SGMAEs, and DSCA-Net improve regional integrity, certain category confusion remains in boundary transition areas and local small regions. In contrast, the classification map of ABCL is closer to the ground truth, with more coherent homogeneous regions and more complete boundary contours, and no obvious large-scale confusion between known classes and unknown regions is observed. In the red dashed boxes in Fig. 6, ABCL can accurately identify known classes and reject unknown classes, and the interclass boundaries are more accurately and

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TABLE III
CLASSIFICATION RESULTS (%) OF DIFFERENT METHODS ON THE LN02 DATASET

Class	10 Training Samples Per Class							20 Training Samples Per Class						
	SpectralFormer	SSFTT	MDL4OW	SSMLP-RPL	SGMAEs	DSCA-Net	ABCL	SpectralFormer	SSFTT	MDL4OW	SSMLP-RPL	SGMAEs	DSCA-Net	ABCL
1	88.67	94.64	95.82	83.62	92.86	87.99	**97.18**	96.24	98.43	96.19	96.34	96.62	90.90	**99.58**
2	98.97	89.40	**99.57**	86.21	79.62	91.93	96.37	99.25	95.73	**99.86**	97.35	90.76	98.24	97.33
3	91.18	**96.53**	95.78	85.36	91.75	86.56	95.96	98.96	98.91	**99.47**	95.72	95.68	94.42	97.04
4	95.15	94.51	**95.46**	89.56	90.66	85.51	91.51	97.34	**97.54**	96.42	92.97	93.85	91.41	93.35
5	94.69	97.46	**99.12**	92.51	94.10	89.36	96.72	96.39	98.84	**99.15**	97.91	98.08	98.20	98.13
6	92.33	93.20	74.21	89.10	**95.71**	95.21	95.16	93.00	97.74	86.67	90.31	99.38	**99.46**	98.90
7	78.74	**99.19**	84.95	84.05	97.57	89.46	95.23	88.34	**100.00**	90.10	95.84	96.04	**100.00**	96.50
9	76.84	95.57	87.49	82.71	93.40	93.35	**95.92**	88.28	**97.12**	89.23	93.13	96.66	96.29	96.99
Unknown	6.17	1.51	10.07	37.68	7.89	3.36	**46.57**	2.97	1.76	9.88	29.10	5.71	2.69	**42.82**
OA	85.63±2.68	92.09±2.80	89.53±0.87	84.25±4.61	87.72±3.31	87.57±4.68	**96.15±0.92**	92.36±1.15	95.94±0.74	92.54±0.73	93.58±1.85	93.18±1.62	92.72±3.04	**97.78±0.71**
AA	80.30±2.57	84.67±1.38	82.50±0.94	81.20±2.96	82.62±1.53	80.30±3.21	**90.07±1.22**	84.53±1.30	87.34±0.64	85.22±1.01	87.63±1.33	85.86±0.98	85.73±1.39	**91.18±1.16**
Kappa	82.20±3.36	90.08±3.40	88.06±1.08	80.79±3.28	84.76±3.92	84.74±3.49	**95.12±1.83**	91.10±1.45	94.86±0.92	91.33±0.91	92.17±2.28	91.41±1.99	90.86±2.68	**97.18±0.59**

smoothly delineated. Combined with the quantitative results, ABCL also achieves the best overall performance on the SA dataset, indicating that the proposed APT, AEM, and CRM can continue to improve spatial consistency and open-set discriminative ability in relatively regular scenes.

3) Results Analysis on the LN02 Dataset: Table III presents the classification results of different methods on the LN02 dataset. Compared with the IP and SA datasets, LN02 contains more complex wetland scenes with abundant spatial structures, posing greater challenges for open-set semi-supervised classification. Under the setting with ten labeled samples per known class, ABCL achieves 46.57%, 96.15%, 90.07%, and 95.12% in terms of unknown, OA, AA, and Kappa, respectively, outperforming all compared methods. From the class-wise perspective, ABCL consistently achieves superior performance on most categories while maintaining the highest unknown-class recognition accuracy. Compared with closed-set methods, such as SpectralFormer, SSFTT, ABCL exhibits significantly stronger unknown-class discrimination capability. Although MDL4OW, SSMLP-RPL, SGMAEs, and DSCA-Net introduce open-set or semi-supervised learning strategies, they still suffer from unknown-class interference. Benefiting from the joint optimization of APT, AEM, and CRM, ABCL effectively improves feature discrimination and adaptive boundary modeling.

When the number of labeled samples per known class increases to 20, the performance of all methods is further improved. Under this setting, ABCL achieves 42.82%, 97.78%, 91.18%, and 97.18% in terms of unknown, OA, AA, and Kappa, respectively, and still achieves the best overall performance among all compared methods. These results indicate that ABCL can simultaneously improve the recognition accuracy of known classes and maintain robust unknown-class rejection as more labeled samples become available, further demonstrating the effectiveness and robustness of the proposed method on complex wetland scenes.

Fig. 7 shows the classification maps of different methods on the LN02 dataset. Compared with the IP and SA datasets, the LN02 dataset contains more complex wetland scenes with diverse land-cover categories and intricate boundary structures. As shown in the figure, SpectralFormer and SSFTT produce noticeable scattered misclassified pixels, while MDL4OW, SSMLP-RPL, SGMAEs, and DSCA-Net improve the regional consistency but still exhibit category confusion around complex boundaries and unknown regions. In contrast, ABCL generates classification maps that are closer to the ground truth, with smoother homogeneous regions and more accurate boundary delineation. As highlighted by the red dashed boxes in Fig. 7, ABCL more accurately distinguishes known classes from unknown regions while preserving finer spatial details. These visual results are consistent with the quantitative comparisons in Table III, further demonstrating that the proposed APT, AEM, and CRM effectively enhance spatial consistency and open-set discriminative capability on complex wetland scenes.

TABLE IV
IMPACT OF THE PROPOSED APT ON CLASSIFICATION PERFORMANCE (%)

Dataset	PE	APT	Unknown	OA	AA	Kappa
IP	✓		47.56	74.65	86.23	70.62
		✓	58.24	79.56	87.65	76.28
SA	✓		30.76	83.19	87.77	81.36
		✓	37.35	88.20	91.14	86.87
LN02	✓		39.15	90.67	85.24	89.93
		✓	46.57	96.15	90.07	95.12

D. Ablation Experiments

In this section, we conduct systematic ablation experiments to analyze in depth the impact of each component of the ABCL model on the HSI classification performance.

1) Impact of the Proposed APT: The proposed APT explicitly decouples the positional modeling into horizontal and vertical components, introducing direction-aware spatial biases into the self-attention mechanism. This design enables the model to more finely perceive the anisotropic spatial structures inherent in HSIs, thereby improving its ability to distinguish boundary regions and reducing the risk of over-smoothing in open-set scenarios where unknown classes may be present. To validate the effectiveness of APT, we compare it with conventional PE under the same experimental settings. As shown in Table IV, replacing APT with PE leads to consistent performance degradation on all three datasets (using ten labeled samples per known class). On the IP dataset, which exhibits complex spatial boundaries and strong category

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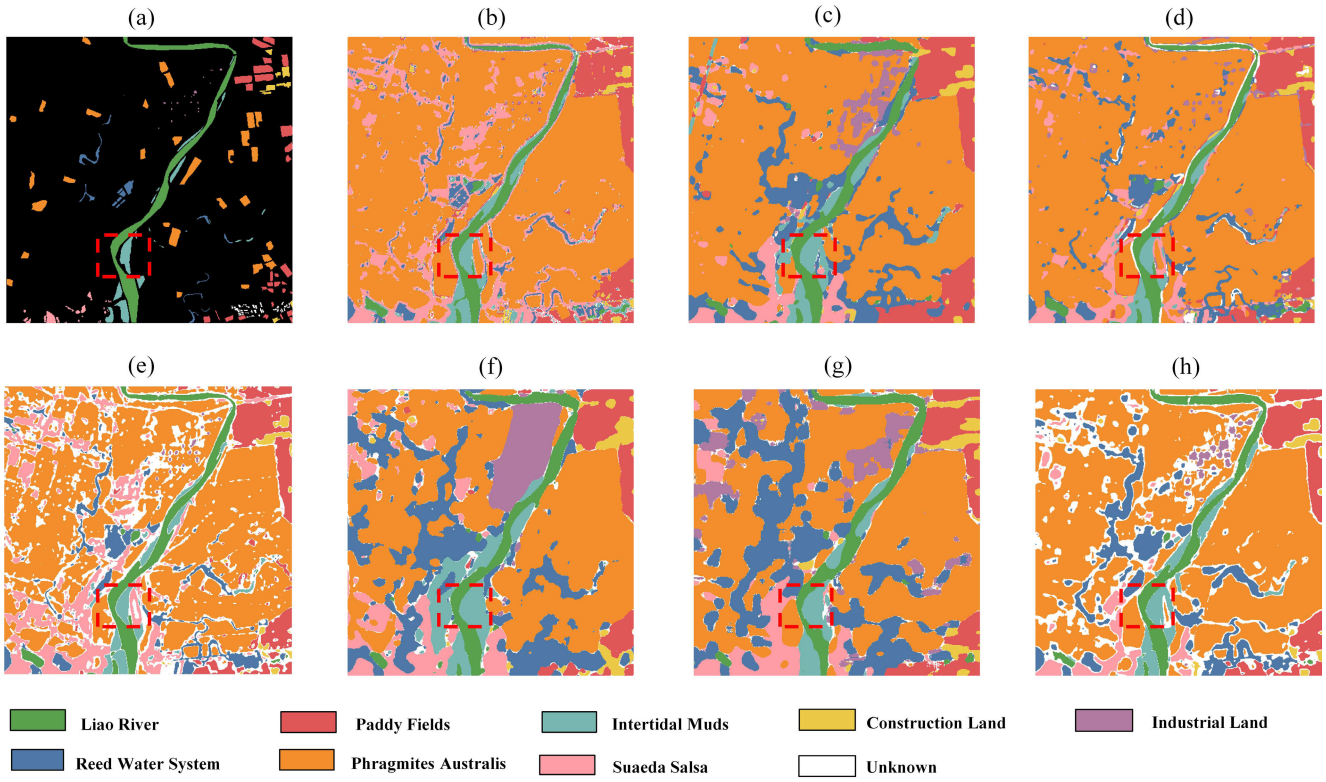


Fig. 7. Classification maps of different methods on the LN02 dataset. (a) Ground truth. (b) SpectralFormer. (c) SSFTT. (d) MDL4OW. (e) SSMLP-RPL. (f) SGMAEs. (g) DSCA-Net. (h) Proposed ABCL.

mixing, the absence of APT reduces Unknown accuracy by 10.68% and OA by 4.91%. On the SA dataset, which is relatively regular but still contains challenging categories, the corresponding decreases are 6.59% and 5.01%, respectively. On the LN02 dataset, which contains more complex wetland scenes with abundant spatial structures, removing APT decreases the Unknown accuracy and OA by 7.42% and 5.48%, respectively. These results demonstrate that APT's direction-sensitive spatial modeling not only enhances the discrimination of known classes but also significantly improves the model's ability to reject unknown-class samples, particularly in areas with complex boundaries or high interclass similarity. The improvements are consistent with the motivation stated in the introduction: by capturing anisotropic spatial priors, APT sharpens category boundaries and suppresses oversmoothing, which is essential for open-set semi-supervised HSI classification.

2) Impact of the Proposed AEM: AEM effectively alleviates the interference of unknown-class samples on consistency learning by performing uncertainty-aware adaptive weighting on unlabeled samples, and further enhances the model's ability to characterize open-set discriminative boundaries. To validate the effectiveness of AEM, we compare the full ABCL model with a variant that replaces AEM with a fixed-confidence hard selection strategy (denoted as "ABCL w/o AEM"), under the setting of ten labeled samples per known class. Table V reports the results on all three datasets. On the IP dataset, which features complex spatial boundaries and strong category mixing, the introduction of AEM improves unknown accuracy by 1.82% and OA by 3.04%. On the SA dataset, which is relatively more regular but still contains challenging categories, AEM yields improvements of 1.27% in unknown and 2.56% in OA. On the LN02 dataset, AEM further improves the unknown accuracy and OA by 4.19% and 2.55%, respectively. These results demonstrate that AEM not only enhances the model's ability to reject unknown-class samples but also maintains or even improves the classification accuracy of known classes. The gains are particularly notable on the IP dataset, where the interplay between complex boundaries and unknown-class interference is more severe, highlighting AEM's role in adaptively calibrating the contribution of uncertain samples. Overall, the proposed AEM effectively mitigates the negative impact of unknown-class interference on consistency learning, which aligns with the motivations stated in the introduction and abstract: reducing unknown-class interference and improving feature discriminability for open-set scenarios.

TABLE V
IMPACT OF THE PROPOSED AEM ON CLASSIFICATION PERFORMANCE (%)

Dataset	Method	Unknown	OA	AA	Kappa
IP	ABCL w/o AEM	56.42	76.52	87.59	72.86
	ABCL w/ AEM	58.24	79.56	87.65	76.28
SA	ABCL w/o AEM	36.08	85.64	89.31	84.07
	ABCL w/ AEM	37.35	88.20	91.14	86.87
LN02	ABCL w/o AEM	42.38	93.60	88.72	92.35
	ABCL w/ AEM	46.57	96.15	90.07	95.12

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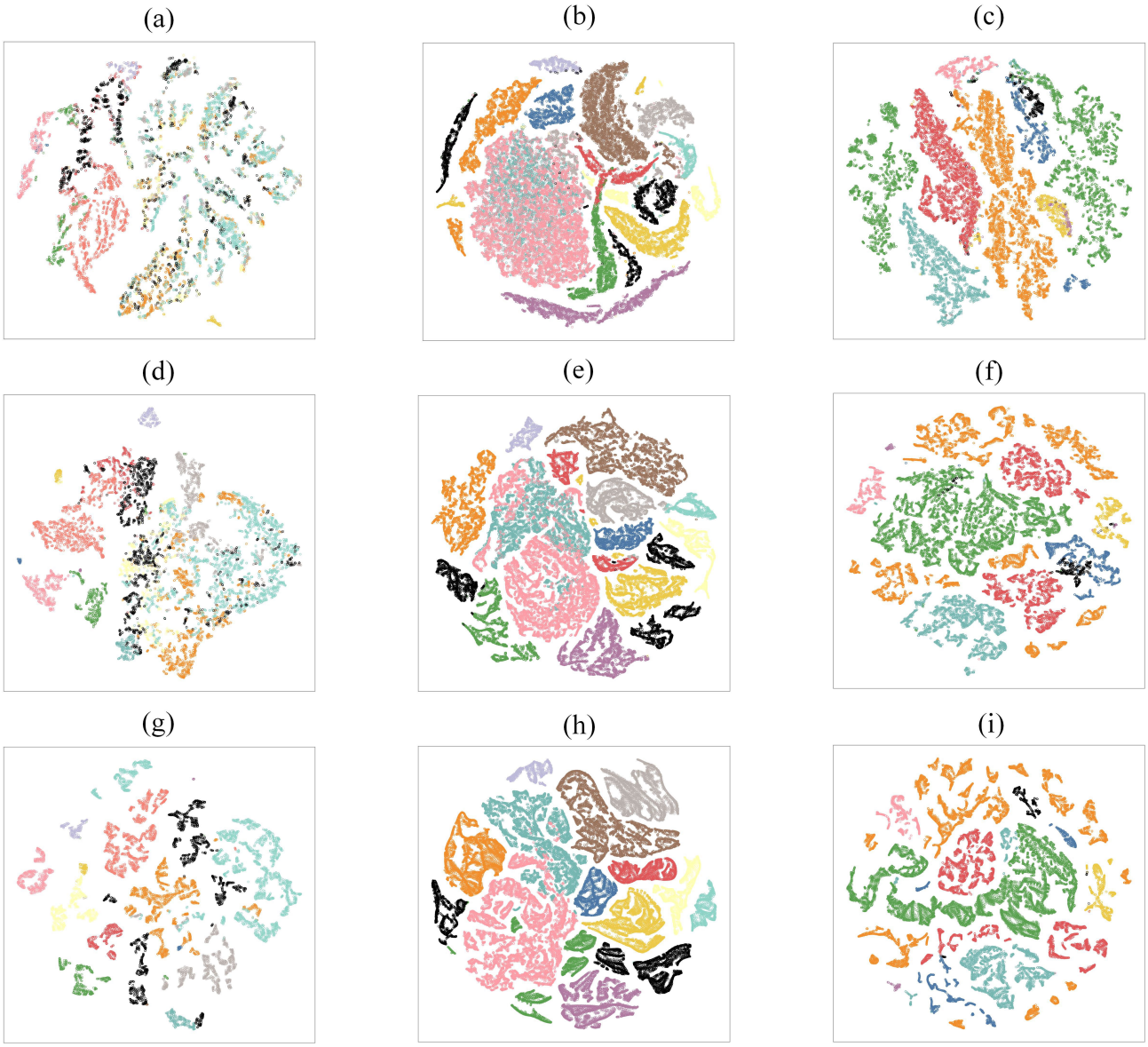


Fig. 8. t-SNE visualization results of feature space. (a) Original feature space on IP. (b) Original feature space on SA. (c) Original feature space on LN02. (d) Feature space constructed by SSMLP-RPL on IP. (e) Feature space constructed by SSMLP-RPL on SA. (f) Feature space constructed by SSMLP-RPL on LN02. (g) Feature space constructed by ABCL on IP. (h) Feature space constructed by ABCL on SA. (i) Feature space constructed by ABCL on LN02.

TABLE VI
IMPACT OF THE PROPOSED CRM ON CLASSIFICATION PERFORMANCE (%)

Dataset	Method	Unknown	OA	AA	Kappa
IP	ABCL w/o CRM	38.65	78.36	87.14	75.78
	ABCL w/ CRM	58.24	79.56	87.65	76.28
SA	ABCL w/o CRM	25.18	87.41	90.81	85.08
	ABCL w/ CRM	37.35	88.20	91.14	86.87
LN02	ABCL w/o CRM	31.23	95.70	89.53	94.56
	ABCL w/ CRM	46.57	96.15	90.07	95.12

3) Impact of the Proposed CRM: The proposed CRM constructs class prototypes from labeled features and enforces prototype-based constraints (10)–(12), explicitly pulling features of the same class toward their corresponding prototype while pushing apart features from different classes. This design aims to enhance intraclass compactness and interclass separability, thereby providing a more structured and discriminative feature space for open-set discrimination, as motivated in the introduction and abstract. To evaluate the contribution of CRM, we compare the full ABCL model with a variant that removes CRM (denoted as "ABCL w/o CRM"), under the setting of ten labeled samples per known class. Table VI reports the results on all three datasets. On the IP dataset, incorporating CRM improves unknown accuracy by 19.59% and OA by 1.20%. On the SA dataset, the corresponding improvements are 12.17% for unknown and 0.79% for OA. On the LN02 dataset, CRM further improves the unknown accuracy and OA by 15.34% and 0.45%, respectively. The substantial gains in unknown accuracy demonstrate that CRM significantly enhances the model's ability to reject

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unknown-class samples by enforcing a more separable feature space. The relatively smaller improvements in OA are expected, as the baseline ABCL without CRM already achieves competitive known-class classification accuracy; CRM primarily refines the feature space structure without compromising the recognition of known classes. The larger improvement on the IP dataset compared with SA is consistent with the fact that IP exhibits more severe intraclass variability and interclass similarity, making the feature space more amenable to prototype-based regularization. Overall, these results confirm that CRM serves as a critical component for improving open-set classification performance, aligning with its motivation of providing a stable representation basis for open-set discrimination by enhancing intraclass compactness and interclass separability.

E. Analysis of Feature Space Visualization

To intuitively evaluate the feature representation capability of different methods, t-SNE is employed to visualize the original features and the feature spaces learned by SSMLP-RPL and ABCL on the IP, SA, and LN02 datasets, as shown in Fig. 8. The category colors are consistent across different methods, and the unknown class is represented in black. The original features in Fig. 8(a)–(c) exhibit scattered distributions and evident overlap among different classes. After feature learning with SSMLP-RPL, the distributions become more structured, but noticeable intraclass dispersion and interclass confusion still remain.

In contrast, the feature spaces constructed by ABCL in Fig. 8(g)–(i) show more compact intraclass distributions and clearer interclass boundaries. Most unknown samples are also better separated from the known-class clusters. These results indicate that ABCL can construct a more discriminative feature space by enhancing intraclass compactness and interclass separability, which is consistent with the joint effects of APT, AEM, and CRM.

F. Parameter Analysis

1) Impact of Spatial Size: In HSI classification, the spatial size of the input neighborhood patch determines the amount of contextual information available for predicting the center pixel. A larger patch provides more neighboring pixels, which can supply richer spectral–spatial cues; however, excessively large patches may introduce pixels from other land-cover classes, thereby contaminating the feature representation and degrading classification accuracy. This tradeoff is particularly critical in open-set semi-supervised scenarios, where unknown-class samples may lie near the boundaries of known classes. To investigate the effect of spatial size on the proposed ABCL, we conducted experiments with patch sizes ranging from 5×5–21×21 under the setting of ten labeled samples per class, as shown in Fig. 9. On the IP and SA datasets, the classification accuracy first increases and then gradually decreases as the spatial size grows, with the best performance achieved at 15×15. In contrast, the LN02 dataset reaches its highest accuracy at a smaller spatial size of 9×9, after which the performance gradually declines as larger neighborhoods introduce more heterogeneous contextual information. The performance drop beyond the optimal spatial size is more pronounced on the IP and LN02 datasets, both of which contain more complex spatial structures and stronger local heterogeneity. This observation confirms that excessively large neighborhoods are more likely to introduce pixels from adjacent categories, thereby increasing interclass confusion and reducing classification accuracy. Therefore, based on the above observations, a spatial size of 15×15 is adopted as the default setting for the IP and SA datasets, while a spatial size of 9×9 is adopted for the LN02 dataset.

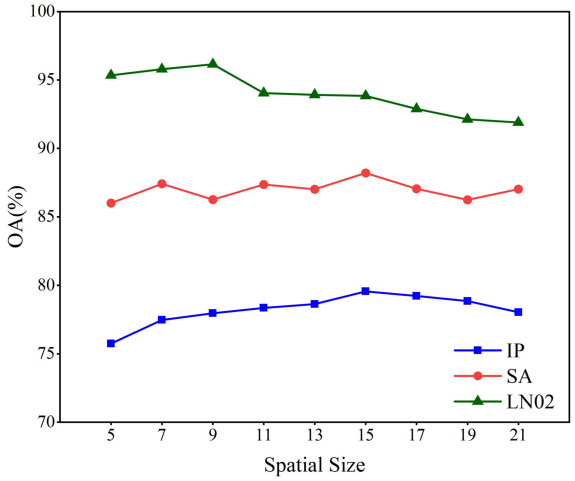


Fig. 9. Classification performance of ABCL under different spatial sizes (%).

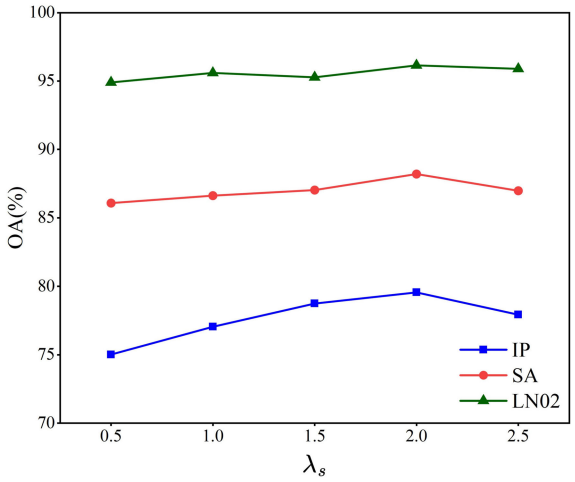


Fig. 10. Classification performance of ABCL under different soft-weight parameters (%).

2) Impact of Soft-Weight Parameter of AEM (λ_s): The AEM dynamically adjusts the contribution of each unlabeled sample to the consistency loss based on its predictive uncertainty. Specifically, the soft-weight parameter λ_s controls the decay intensity when the normalized entropy is mapped to sample weights, thereby regulating how much high-uncertainty (likely unknown-class) samples are suppressed. To identify the optimal λ_s, we conducted experiments with values ranging from 0.5 to 2.5 under the ten labeled samples per class setting (Fig. 10). On all three datasets, the best classification

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performance is achieved at $\lambda_s = 2.0$. The SA dataset, which is relatively more regular, shows gradual improvement from 0.5 to 2.0 and then a decline at 2.5, while the IP dataset exhibits a sharper performance peak at 2.0 due to its higher interclass similarity and boundary complexity. The LN02 dataset follows a similar trend, achieving the highest classification accuracy at $\lambda_s = 2.0$. These results indicate that a moderate soft weighting ($\lambda_s = 2.0$) effectively balances the exploitation of reliable unlabeled samples and the suppression of uncertain unknown-class interference, aligning with the design motivation of AEM to adaptively calibrate consistency learning in open-set environments. Therefore, $\lambda_s = 2.0$ is adopted as the default setting.

V. Conclusion

In this study, we propose an ABCL model. The proposed method achieves low-cost and high-accuracy perception while simultaneously considering known-class classification and unknown-class rejection. Specifically, we first design an APT, which introduces direction-sensitive spatial priors by decoupling horizontal and vertical positional biases, thereby enhancing the model's representation capability for complex land-cover boundaries and spatial structural differences. Second, an AEM is proposed to guide consistency learning, thereby improving the discriminability of model features. In addition, a CRM is introduced to further optimize the feature space structure by enhancing intraclass compactness and interclass separability, providing a more stable representation basis for open-set discrimination. Extensive experiments on the IP, SA, and LN02 datasets demonstrate that the proposed ABCL achieves excellent performance on both known and unknown classes, significantly outperforming other advanced HSI classification methods. From a practical perspective, ABCL has potential applications in dynamic land-cover mapping, environmental monitoring, and precision agriculture, where limited labeled data may not cover all categories encountered during deployment. Its ability to recognize known categories while rejecting previously unseen ones may help improve the reliability of hyperspectral interpretation in such dynamically changing scenarios.

Despite its effectiveness, ABCL still has some aspects that can be further explored. First, the consistency learning strategy in AEM is designed based on the uncertainty estimation of weak-view predictions, which provides an effective way to adaptively adjust the contribution of unlabeled samples. However, more comprehensive uncertainty modeling strategies that incorporate richer semantic cues or complementary views may further improve the consistency learning process in complex open-set scenarios. Second, the current experiments are mainly conducted under predefined open-set settings on benchmark datasets. Further investigations under different openness levels, semantically similar unknown classes, and cross-scene or cross-sensor conditions would provide a more comprehensive evaluation of the generalization capability of ABCL. Future work will focus on exploring more advanced consistency learning strategies and broader open-set adaptation mechanisms to further enhance the robustness of the proposed framework.

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